- Electro-optical/infrared (EO/IR) turret for day/night surveillance and targeting.
- Synthetic aperture radar (SAR) for ground mapping and target tracking.
- Laser designator/range finder for precision-guided munitions.
- Data link communication system with line-of-sight (LOS) and satellite control options.

The UAV's onboard mission management computer integrates multi-sensor data in real time, allowing autonomous flight and target acquisition. The ground control station (GCS) features dual-console operator interfaces for flight and payload control, with live video feeds transmitted via encrypted links.

Armament and combat capability

The Wing Loong I is a multi-role armed UAV, capable of carrying up to 200 kg of ordnance across four underwing hardpoints. Its weapons suite includes:

- BA-7 air-to-surface missiles (equivalent to AGM-114 Hellfire).
- GB3 laser-guided bombs.
- Blue Arrow-7 anti-armor missiles.
- LS-6 precision-guided bombs.

These systems allow the UAV to conduct close air support (CAS), counterterrorism, and anti-insurgency missions with precision.

Operational applications

Since its induction, the Wing Loong I has been deployed by several countries, including Saudi Arabia, Egypt, the United Arab Emirates,